

Viktor Nikoriak

Python GIS Engineer | Hydrologist | GeoAI Researcher

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Building reproducible geospatial data systems for flood-risk modeling, DEM validation and AI-assisted environmental analytics.

Professional Summary

Python GIS Engineer & Hydrologist with 15+ years of domain experience and a focus on AI-driven environmental analytics, floodplain modeling and DEM validation. Founder of **GeoHydroAI** and recipient of the **DBU Environmental Innovation Grant**, with proven expertise in building automated geospatial data pipelines using Python, PostGIS, DuckDB and remote sensing data.

My work connects satellite imagery, ICESat-2 LiDAR altimetry, hydrological and hydraulic modeling, and interactive dashboards to support flood-risk assessment, environmental monitoring and spatial data-quality validation.

Core Competencies

- Geospatial Data Engineering
- Remote Sensing & DEM / LiDAR Validation
- Floodplain Modeling & Flood-Risk Assessment
- Python GIS Pipelines (GeoPandas, GDAL, Rasterio, xDEM)
- Spatial Databases & Analytics (PostGIS, DuckDB)
- Interactive Dashboards & Scientific Communication

Research & Engineering Focus

- Geospatial data pipelines
- Satellite data processing
- Large-scale environmental datasets
- Spatial data quality evaluation
- Map data validation and matching
- AI and machine learning for geospatial analytics

Technical Skills

Programming

- Python
- R
- SQL
- TypeScript

Geospatial Python

- GeoPandas
- Shapely
- Rasterio
- xDEM

- PyProj
- GeoParquet

Remote Sensing & GIS

- GDAL
- ICESat-2, Sentinel-1/2, Landsat
- Google Earth Engine
- SAGA GIS, GRASS GIS
- QGIS, ArcGIS, ArcGIS Online

Data Engineering

- ETL / ELT pipelines
- Big geospatial datasets processing
- Spatial indexing
- Data quality pipelines

Databases

- PostgreSQL / PostGIS
- DuckDB
- MongoDB
- SQLite
- Redis

Cloud & Infrastructure

- AWS
- Docker
- Linux
- Git

AI / Machine Learning

- Random Forest
- MLP
- CatBoost
- DEM error modelling
- Geospatial machine learning

LLM & AI Pipelines

- Retrieval-Augmented Generation (RAG)
- LLM data pipelines
- AI-driven geospatial analytics

Visualization

- Plotly
- Dash
- Matplotlib
- Streamlit
- Deck.gl

Selected Projects

GeoHydroAI Platform

Open geospatial platform integrating satellite data, DEM validation, hydrological models and machine learning for flood-risk assessment, environmental monitoring and landscape-restoration analytics.

Includes interactive flood scenario visualization dashboards:

geohydroai.org/flood_scenarios/

Technologies: Python, GeoPandas, Dash, DuckDB, PostGIS, Docker, TypeScript, React, Deck.gl.

Sentinel-1 Flood Detection Pipeline

Automated SAR flood detection pipeline using Sentinel-1 imagery, SNAP processing and Python microservices. The workflow covers preprocessing, terrain correction and water classification for operational flood mapping experiments.

Python Course Repository

Educational Python repository with practical exercises, homework examples and GitHub-based learning workflows for beginner students.

Repository: [Python Course GitHub Repository](#)

Professional Experience

Ukrainian Hydrometeorological Institute (UHMI)

Junior Researcher / PhD Student

2022 – Present

Kyiv, Ukraine

- Engineered automated Python pipelines for processing Sentinel-1/2 imagery and global DEMs, improving reproducibility of flood-risk and DEM-validation workflows.
- Built geospatial validation workflows combining ICESat-2 LiDAR altimetry, vertical reference transformation, xDEM, GDAL, GeoPandas and PostGIS to assess DEM accuracy in mountainous floodplain environments.
- Led validation of global elevation models including SRTM v3, NASADEM, ASTER GDEM v2, ALOS AW3D30, Copernicus GLO-30, FABDEM and TanDEM-X for floodplain modeling in the Ukrainian Carpathians.
- Developed flood-modeling workflows for the Prut River Basin using HEC-HMS and HEC-RAS 2D, supporting scenario-based flood hazard assessment.
- Designed spatial data-quality pipelines with Python, PostGIS and DuckDB for large raster-vector datasets used in environmental monitoring.
- Created interactive analytical dashboards with Dash and Plotly to communicate geospatial results and support exploratory analysis.

Chernivtsi Regional Hydrometeorological Center

Hydrologist

2006 – 2022

- Conducted river discharge measurements using ADCP instruments and performed GNSS-based river cross-section surveys for operational hydrological monitoring.
- Supported hydrological forecasting workflows by collecting, validating and interpreting field observations from river monitoring stations.
- Contributed to EU flood-protection projects through field measurements, hydrometric data preparation and technical support for flood-risk analysis.
- Built long-term domain expertise in river hydrology, flood processes and operational water-level monitoring in the Prut and Siret river basins.

Python Instructor – Beetroot Academy

2026 – Present

- Teach introductory Python programming to beginner students through practical examples, homework assignments and project-based exercises.
- Design educational materials explaining variables, loops, functions, data structures and basic algorithms in a clear step-by-step way.

- Introduce students to Git, GitHub and collaborative development workflows used in real software projects.
- Mentor students through debugging, code review and gradual improvement of programming thinking.

Grants and Awards

DBU Environmental Innovation Grant

GeoHydroAI – Data-driven Flood Risk and Landscape Restoration for Protecting Natural and Cultural Heritage in Post-war Ukraine

Awarded by the German Federal Environmental Foundation (DBU).

- Integration of ICESat-2 satellite altimetry
- Machine learning DEM validation
- Hydrological modelling with HEC-HMS
- Hydraulic flood modelling with HEC-RAS

Partner: Bavarian Environment Agency (LfU), Germany

Education

PhD Student – Hydrology and Water Resources

Ukrainian Hydrometeorological Institute
2023 – Present

MSc – Ecology and Environmental Protection

Chernivtsi National University
1999 – 2004

Publications

Validation of global elevation models using ICESat-2 LiDAR data for floodplain modeling in the Ukrainian Carpathians

Authors: V. V. Nikoriak, V. V. Osyrov, N. M. Osadcha
Journal: Geofizicheskiy Zhurnal, 48(2), 2026
DOI: [10.24028/gj.v48i2.346831](https://doi.org/10.24028/gj.v48i2.346831)

A High-Resolution Hydrological Dataset for Ukrainian River Basins With an Interactive Web Interface

Journal: Geoscience Data Journal, 2025
DOI: [10.1002/gdj3.70027](https://doi.org/10.1002/gdj3.70027)

Certifications and Workshops

- **USACE / UNESCO – Hydraulic Modeling with HEC-RAS** (2025)
- **Summer Academy – Bavarian Environment Agency** (2025)
- **Data Science Course – Sigma Software University** (2025)
- **SWAT+ Advanced Workshop – Texas A&M** (2025)
- **Rule of Law Ideals – CEU** (2025)
- **Academic English (2 ECTS) – CEU** (2025)
- **Research Methods for the Social Sciences – CEU** (2025)
- **Erasmus+ CLUVEX Virtual Exchange Week** (2025)
- **Processing and Analysis of Satellite Images on GEE – JASU** (2025)
- **SWAT+ Beginner Workshop – Texas A&M** (2024)
- **Remote Sensing – Junior Academy of Sciences of Ukraine** (2024)
- **Introduction to Cloud Processing of Sentinel Data – Copernicus** (2024)

- [HEC-HMS and HEC-RAS Workshop – LfU & TUM](#) (2024)
- [Python Development Course – Beetroot Academy](#) (2021)

Languages

Ukrainian – Native

English – Upper-Intermediate (B2)